

Permafrost active-layer thickness prediction under sparse observations using physics-based pseudo-label augmentation and machine learning

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Abstract

We provide a label-count workflow for mapping active-layer thickness in sparsely measured permafrost regions with the Stefan model and machine learning (ML). ML models are mostly tested within training regions at one label count, without recalibrated baselines. We held out five regions with 100 km buffers and compared root-mean-square error (RMSE) from zero to all labels against Stefan models with source and recalibrated coefficients (internally pre-registered). Without labels, Stefan pseudo-labels gave lower error than shuffled ones (-1.6 cm; 95% confidence interval -2.0 to -1.0); with ten labels, recalibration gave lower four-region mean error than the source coefficient (-2.5 cm; -3.0 to -1.9 ; lower in one region), with no further difference established for residual ML. In analyses designed after these results, method selection by within-target cross-validation with 40 and 160 labels changed RMSE by -0.87 and -1.3 cm against a fixed residual recipe (two regions, change from Canada). Within Alaska, residual ML reached 13.9 against 14.4 cm RMSE for recalibrated Stefan (1000 labels); about 99% of the squared-error reduction (all labels) lay between climate grid cells. Spreading 100 labels over blocks changed RMSE against random sampling by -2.5 cm (Canada) and $+1.7$ cm (Lena Delta). Sub-grid covariates are needed next. [PENDING: XE-a, XE-b, satellite inputs (stage 2); registered deadline 8 October 2026, 14:22 KST]

Keywords: Active-layer thickness, Permafrost, Stefan model, Physics-guided machine learning, Spatial transfer, Sampling design

Introduction

Active-layer thickness (ALT) is the maximum seasonal thaw depth of the ground above permafrost, and its change affects infrastructure built on permafrost^{1,2}. ALT is measured at a limited number of sites, many of them in the 35-year Circumpolar Active Layer Monitoring (CALM) compilation³. In the five regions evaluated here, the number of measured ALT values (labels) ranges from 30 in E Russia to 13,606 in Alaska (Fig. 1a; Table 1). The Stefan solution⁴ predicts ALT as a coefficient E times the square root of the thawing degree-days (TDD), with E calibrated by local thaw-depth measurements⁵. The geometric mean of E over labelled cells ranges from $1.31 \text{ cm } (^{\circ}\text{C d})^{-1/2}$ in the Lena Delta to 11.03 on the Tibetan Plateau (Fig. 1b). Because measurements in a new permafrost region (the target region) are few, the error of ALT maps there depends on models fitted to other regions (the source data).

Outputs of the Stefan or Kudryavtsev models have served as machine-learning (ML) input features^{6,7}, E has been predicted from covariates^{2,8}, labels generated by physical rules have trained a zero-curtain detector⁹, and process-model pre-training and residual modelling are used in other environmental fields^{10–12}. Gautam et al. compared a random forest with a Stefan model at 68 Alaskan CALM sites with one random 70/30 split (test coefficients of determination 0.24 and 0.54)¹³. Large-scale ALT maps have been validated with distance blocks^{1,14,15}. For clustered spatial data, random cross-validation can overstate map accuracy^{16–18}, and unbiased map accuracy has been argued to require probability sampling with design-based inference rather than spatial cross-validation¹⁹. The ALT evaluations above each use one label condition, whereas adjacent fields have evaluated models in held-out regions or with few local observations, in some cases against calibrated process models, for lake temperature^{10,11,20}, ungauged catchments^{21,22}, subsurface temperature²³ and spatial risk prediction²⁴. To our knowledge, no study of permafrost ALT or mean annual ground temperature has reported the error in regions excluded from the training data as a function of the number of target labels, relative to a Stefan model recalibrated with those labels.

A new region needs a rule for combining the Stefan model and ML as target labels accumulate, and a procedure for adding labels. We address three questions. (1) When no target labels are available, what does physics information add to ML, and where does the gain come from? (2) As target labels increase from ten to thousands, how much does ML add to a Stefan model recalibrated with the same labels, and at which spatial scale does the gain arise? (3) As labels accumulate, how should the method be selected, and by which procedure should new labels be placed?

Each target region was held out from the source data with a 100 km buffer, labels were drawn from half of its 0.5° blocks, and errors were scored on the other half, for 0, 3, 10, 40, 160, 320, 1000 and all available labels (Fig. 1c,d). Within label-rich regions, block-holdout tests designed after earlier results were viewed measured the gain of ML over the recalibrated model. Contrasts were judged from block-bootstrap 95% confidence intervals (CIs) with a $\pm 0.5 \text{ cm}$ equivalence margin, and the transfer analysis was internally pre-registered. We find that, without target labels, Stefan pseudo-labels (Stefan predictions used as training labels) reduced the four-region mean root-mean-square error (RMSE) relative to shuffled pseudo-labels, that recalibrating the Stefan

coefficient with ten labels reduced it relative to the source coefficient (lower error in one of the four regions), and that within Alaska, with all labels, almost all of the gain of ML over the recalibrated model came from correcting bias between climate grid cells. The contribution is the evaluation design, its results including negative ones, and the label-count workflow built from them.

Results

Evaluation design and validation ladder

On the holdout protocol described above (Fig. 1c,d), every method was compared with two Stefan baselines that used the same target labels (Fig. 2a,b): the source-coefficient Stefan model, whose coefficient E_0 was estimated from the source cells (Table 1), and the Stefan model recalibrated with the target labels by shrinkage towards E_0 ($\kappa = 10$)²⁵. Four-region means weight the Lena Delta, Canada, W Russia and E Russia equally and exclude Alaska, which is a held-out reference target. Intervals are cell-weighted unless marked block-equal.

Averaged over five regions with equal weight, the RMSE of direct ML increased from 22.9 cm under random cell splits to 33.3 cm under region holdout (difference 10.3 cm; 95% confidence interval (CI) 7.3 to 12.8; block-equal 6.1 cm, 4.7 to 7.6), whereas a Stefan model with the coefficient fitted by least squares to the training cells stayed between 26.8 and 27.0 cm (Fig. 1e). The 10.3 cm increase is reported as an effect size rather than a confirmatory result, and the registered contrast between cluster holdout with a 500 km buffer and random cells was 8.2 cm (95% CI 5.6 to 10.4; Supplementary Table S1). Moving from random-cell to k -fold nearest-neighbour distance matching (kNNDM) splits²⁶ increased the RMSE of direct ML by 5.76, 8.17 and 14.20 cm and that of the Stefan model fitted to the training cells by 0.29, 0.51 and 4.28 cm in Alaska, the Lena Delta and Canada (cell-weighted; descriptive, designed after earlier results were viewed; Fig. 1e; Supplementary Table S2). Random validation has been argued to estimate map accuracy validly for probability samples¹⁹, but our labels are a clustered non-probability sample, so the errors below are prediction errors on held-out blocks.

Transfer without target labels

Without target labels, pseudo-labels from the Stefan model reduced RMSE relative to shuffled pseudo-labels by 1.6 cm (95% CI 1.0 to 2.0; block-equal 0.6 to 1.5; four regions; Holm-adjusted $P = 0.001$), and in a post hoc count, fewer of 30 targets lost more than 2 cm against the source-coefficient Stefan model with physics pseudo-label augmentation (2 targets) than with direct ML (18 targets) (Fig. 3b; Fig. 4a,b). Against two constant placebos the reductions were 1.7 and 3.6 cm, and against a placebo linear in $\sqrt{\text{TDD}}$ 0.48 cm (registered verdict: mixed). The 2 cm threshold was set after the transfer results had been viewed, the 30 targets include 23 sub-regions or label-expanded versions, and counts based on intervals give the same order (post hoc; Supplementary Table S3).

For the five learners whose primary and common-resampling intervals agreed, direct ML had higher four-region mean error than the source-coefficient Stefan model (Fig. 4c). For the main learner, CatBoost, the difference was 2.2 cm (95% CI 1.1 to 3.4; Holm-adjusted $P = 0.023$), and for two further CatBoost settings, TabPFN v2 and TabICL v2 it was 0.85 to 1.9 cm. These five verdicts depend on the split-independence assumption, and the TabPFN v2 verdict was significant before correction only (Holm-adjusted $P = 0.117$). Residual models with a low weight ($\lambda = 0.25$) on the source-coefficient anchor were equivalent to that model within ± 0.5 cm for three learners (depends on the split-independence assumption).

With Alaska held out, physics pseudo-labels had 1.8 cm higher error than the source-coefficient model (95% CI 0.4 to 2.6). On the Tibetan Plateau, where the source coefficient is far from the regional value (Fig. 1b), direct ML had 60.0 cm lower error (an expected result from an unblinded, cross-environment comparison). With ten labels, direct TabICL v2 had 2.9 cm lower four-region mean error (95% CI 1.1 to 3.6; depends on the split-independence assumption) but 6.0 cm higher error in the three-region mean that includes Alaska.

On the same blocks, after masking blocks within 5 or 25 km of product training sites (registration deviation; designed after earlier results were viewed), the existing CCI v5 map without labels had 23.79 cm higher error than the source-coefficient Stefan model (Holm-adjusted $P = 0.0006$; higher in the Lena Delta, Canada and E Russia), and residual ML had 8.37 and 7.51 cm lower error than CCI v5 recalibrated with the same 10 and all labels (CCI v5 contrasts partly unblinded); no difference was established for the Wei v2 map (47.4% of product values imputed), trained partly on CALM sites in the target regions (ALT definitions differ; Supplementary Table S2).

Coefficient recalibration with three to ten labels

With ten target labels, recalibrating the Stefan coefficient reduced the four-region mean RMSE relative to the source coefficient by 2.5 cm (95% CI 1.9 to 3.0; block-equal 1.9 to 3.3; Holm-adjusted $P = 0.001$), with lower error in one of four regions (W Russia, 12.0 cm), higher error in Canada (2.3 cm; 95% CI 0.7 to 2.9) and no difference established in the Lena Delta and E Russia (regional median change -0.06 cm, $+0.72$ cm without W Russia; Fig. 2c-f; Fig. 3a,d). Adding residual ML to the recalibrated anchor changed RMSE by a further -0.18 cm (95% CI -0.53 to -0.01 ; Holm-adjusted $P = 0.136$), a difference significant before correction only, below 0.5 cm and dependent on the split-independence assumption.

The error reduction of label-based correction with all labels (selection rule of the next section) was rank-correlated with the source-coefficient bias measured with ten labels (Spearman $\rho = 0.38$; 95% CI 0.17 to 0.55; 28 targets; one-sided $P = 0.0002$; designed after the label-grid results had been viewed). The sign of the bias was not related to the reduction ($\rho = -0.01$; Fig. 5a,b; Supplementary Table S4). In a post hoc analysis, no relationship with the size of this bias was established for the recalibration gain, for the share added by the selection rule, or for the ML share beyond recalibration (for the last, positive only with the auxiliary target-fixed interval; the weaker category is reported) (Fig. 5d; Supplementary Table S2).

With ten or fewer labels, the residual structure had lower four-region mean error than ML that receives Stefan outputs as inputs (physics-input ML), by 2.7 cm without labels and 3.7 cm with ten labels (95% CI 2.8 to 5.2; Holm-adjusted $P = 0.001$; with ten labels lower error in two of four regions, regional median change -2.2 cm, -0.32 cm without W Russia) (Fig. 3c). With recalibrated Stefan values as physics inputs, the residual structure still had 2.5 cm lower error but 3.1 cm higher error in Canada, regional verdicts that depend on the split-independence assumption (Supplementary Table S5).

Adding physics pseudo-labels to residual ML was equivalent within ± 0.5 cm with ten labels and increased error by 0.34 cm, below 0.5 cm, with 40 labels in the Lena Delta and Canada (Supplementary Table S5). Label-weighted stacking of physics models and products established no difference from recalibration or residual ML at ten labels, where it fell back to recalibration in 36% of draws (designed after earlier results were viewed, partly unblinded; Supplementary Table S2).

Method selection with tens to hundreds of labels

Selecting the method by five-fold block cross-validation within the target labels reduced RMSE relative to a fixed residual recipe ($\lambda = 0.25$) by 1.3 cm with 160 labels (95% CI 0.7 to 1.7; Lena Delta and Canada; unchanged under common resampling) and was non-inferior at 40 and 160 labels (margin 0.5 cm), with both results arising in Canada; this rule was designed after the label-grid results had been viewed (Fig. 5c; Fig. 6d). With 40 labels, the reduction of 0.87 cm depends on the split-independence assumption. No difference was established in the Lena Delta. With ten labels in the four main regions, non-inferiority was not established, and the rule had 0.70 cm higher error in the Lena Delta; with all labels, non-inferiority depends on the split-independence assumption. The rule did not increase the number of targets with lower error than the recalibrated Stefan model (Supplementary Table S4).

In Canada, no difference from the source-coefficient Stefan model was established (Supplementary Table S4). With Alaska held out, the rule had higher error than the source-coefficient model with 10, 40 and 160 labels (descriptive curve contrasts; Supplementary Table S4).

At 40 and 160 labels, residual ML had 1.9 to 2.4 cm higher error than physics-input ML in the Lena Delta and Canada (the 2.4 cm contrast depends on the split-independence assumption) (Fig. 3c). This difference arose in Canada; with Alaska held out residual ML had lower error. At these label counts no difference was established between residual ML and a stronger recalibrated baseline (the soil-property Stefan model recalibrated with $\kappa = 10$).

Leave-one-source-region-out tuning of neural networks gave learner-specific results, no learner met the registered condition for a learner-general statement, and this tuning was not compared with selection within the target labels (Supplementary Table S4). In new random splits of the same regions (re-test in reused regions, designed after earlier results were viewed, partly unblinded), the sequential workflow had 0.93 cm lower error than random placement with the fixed residual recipe at 160 labels (Lena Delta and Canada; Canada -2.41 cm, Lena Delta $+0.54$ cm, undetermined) and no established difference at 40 labels (minimum detectable effect about 0.5 cm; Lena Delta $+0.21$ cm);

because its placement step reduced to random sampling, this compares the method rule with the fixed recipe on the same labels. In these splits the workflow was non-inferior to the recalibrated Stefan model at 10, 40 and 160 labels (five-region mean, then Lena Delta and Canada), with 1.35 and 1.51 cm lower error at 40 and 160 labels, but only for the pooled mean: it was more than 0.5 cm worse in E Russia at 10 labels (+1.34 cm), in Alaska, the selection family, at 40 and 160 labels (+1.51 and +2.34 cm) and in the Lena Delta at 160 labels (+0.53 cm) (Fig. 6e; Supplementary Table S2). Within regions (auxiliary arm), the rule was equivalent to residual ML at 200 labels, with no difference established at 500 and 1000 labels. Stacking also established no difference at 40 and 160 labels (Lena Delta and Canada; designed after earlier results were viewed, partly unblinded; Supplementary Table S2).

Gains within label-rich regions

Within Alaska, residual ML on the recalibrated Stefan anchor reduced RMSE relative to the Stefan model recalibrated on the same labels by 0.54 cm with 1000 labels (95% CI 0.32 to 0.69; 14.4 to 13.9 cm; unchanged under common resampling), and with all labels about 99% of the reduction in squared error lay between ERA5-Land climate grid cells²⁷ (Fig. 6a,c). This test was designed after earlier results were viewed and registered before running (Methods).

Reductions with 500 and all labels (0.39 and 0.52 cm) depend on the split-independence assumption, and no difference from the source-coefficient Stefan model was established with 1000 or all labels (Supplementary Table S6). In the Lena Delta and Canada, no difference was established for the same contrast at any tested label count (Supplementary Table S6).

With the error split between and within climate grid cells (proxy: 0.5° block and ERA5-Land TDD; Methods; designed after earlier results were viewed and registered before running), residual ML with all labels reduced the between-grid RMSE relative to the recalibrated model by 1.0 cm (95% CI 0.6 to 1.3), and the within-grid RMSE was equivalent within ± 0.5 cm (-0.01 cm; 95% CI -0.03 to 0.04), also under a soil-temperature grid definition (Fig. 6c). Within-grid variation made up 72% of the squared error of the recalibrated Stefan model, of which residual ML explained less than 1% in Alaska and less than 8% in any configuration of the three regions. Attributing the gain to grid-scale bias correction is a post hoc description (error change by scoring block in Fig. 6b).

In the registered within-grid hypothesis, the within-grid predictions of residual ML increased the error in the stratified regional mean with all labels (+0.09 cm; 95% CI 0.02 to 0.40), also under the soil-temperature definition and in Canada, all statistically distinguishable but below 0.5 cm in size (Supplementary Table S7). In transfer, there was no evidence that within-grid predictions matched the observed variation. The registered test of extrapolation to warmer blocks could not be decided (three scoring blocks in Canada; Supplementary Table S7). Three further analyses were designed after earlier results were viewed (Supplementary Table S2). In a spatial proxy with warmer blocks of the same period (extrapolation width 0.97 ($^{\circ}\text{C d}$)^{1/2} in $\sqrt{\text{TDD}}$), which is not a true extrapolation test, residual ML had 2.44 and 2.41 cm lower error than the recalibrated Stefan model at 100 and all labels (non-inferior), and no

difference in extrapolation loss was established between direct ML and either model. Ten terrain, water and tree-cover inputs left the within-grid error of residual ML equivalent within ± 0.5 cm at every label count (auxiliary contrast, partly unblinded). [PENDING: XE-a, XE-b, satellite inputs (stage 2); registered deadline 8 October 2026, 14:22 KST] Stacked residuals had 0.55 and 0.498 cm lower error than recalibrated-anchor residuals at 500 and 1000 labels (two of three regions; the latter below 0.5 cm; not retained when CCI v5 was added), no difference was established at 200 or all labels, and in Alaska stacking was non-inferior to the recalibrated Stefan model (partly unblinded).

Placement of new labels

In Canada, spreading labels over 0.5° blocks or covariate space reduced RMSE relative to random cell sampling by 2.4 cm with 40 transfer labels (95% CI 0.5 to 3.4; covariate max-min distance sampling) and by 2.5 to 3.2 cm with 50 to 100 within-region labels, whereas in the Lena Delta block stratification increased RMSE by 1.7 cm with 100 labels (95% CI 0.9 to 2.4) (Fig. 7a–e). All contrasts used residual ML ($\lambda = 0.25$) on the same splits and were designed after earlier results were viewed and registered before running.

With ten transfer labels, neither strategy had higher error in any of five regions, which is not a non-inferiority result, because one weighting showed higher error for block stratification in the Lena Delta (+1.9 cm) and for covariate spreading in E Russia (+0.36 cm) (Fig. 7f; Supplementary Table S8).

The best Alaskan strategy at 100 labels, covariate-cluster stratification, increased RMSE in the Lena Delta (+0.33 cm, below 0.5 cm). The internally pre-registered contrast of block stratification against random cell sampling, averaged over regions, was undetermined at 10 and 40 labels (two-stage resampling; partly unblinded; same label sets as the transfer contrasts; Supplementary Table S8). We also tested a placement procedure fixed before evaluation (Methods) within the sequential workflow (Fig. 6e). In the new random splits of the same regions this procedure reduced to cell-random sampling, so the placement step contributed nothing and its effect was not tested (Methods); it was chosen in the Alaska family, the cells of the evaluation regions were used to design the placement and the method rule, and the evaluation splits are new partitions of the same cells. Block allocation with covariate coverage established no difference from random placement at 10 or 40 labels, its four settings were non-inferior to block or covariate spreading in 8 of 16 comparisons (2 significant before correction only) and had higher error in 2 (below 0.5 cm), and design-weighted coefficients, a post hoc design aimed at the Lena Delta, increased error in Alaska and Canada (exploratory; designed after earlier results were viewed, partly unblinded; Supplementary Table S2). The candidates were measured cells, so gains are not guaranteed to transfer to field candidates.

Independent regions

In Central Russia, a licence-verified new region with a shallow active layer (57 labels), direct ML without target labels increased RMSE relative to the source-coefficient

Stefan model by 2.7 cm (95% CI 2.1 to 4.1; unchanged under common resampling), and in the five-region shallow pool that includes it, residual ML with all labels reduced RMSE relative to the same model by 3.2 cm (95% CI 2.2 to 4.2; regional median change -0.78 cm, -0.49 cm without W Russia; lower error in W Russia only, higher in Canada) (Table 1). These results carry a cross-environment flag (the pools combine regions fitted on two computing platforms; Methods). No region in the four-region or the five-region pool had lower error with direct ML than with the source-coefficient model at 0 to 40 labels. With ten labels, residual ML and the recalibrated Stefan model were equivalent within ± 0.5 cm in the five-region pool (-0.13 cm; regional median change -0.03 cm), so the four-region result at this label count did not hold there. In Central Russia alone the two weightings disagreed, so no difference was established (Supplementary Table S9).

The Tibetan Plateau, with a deep active layer, is reported separately and not used as independent-region evidence (scored cells outside the source elevation range; unblinded contrasts). There, residual ML with ten labels had a 20.9 cm lower RMSE than the shrunk recalibration but was not better than a least-squares Stefan fit to the same labels (+19.1 cm; Supplementary Table S9). The North Atlantic region, with partly unconfirmed data licences, appears only in Supplementary Table S9. [PENDING: XF, new public regions; registered data deadline 11 October 2026, 14:22 KST; no eligible new macro region found so far] [PENDING: XC-F3, external-family test of the workflow; not tested so far, because placement reduced to random sampling makes its ten-label contrast identical, and no eligible new region; registered sentence after the XF data deadline, 11 October 2026, 14:22 KST] On the Tibetan Plateau, temperature-derived auxiliary rows reduced the transfer error of recalibration with 3 and 10 labels by 8.14 to 83.17 cm (designed after earlier results were viewed, partly unblinded), a result not generalized because these labels differ in definition from direct measurements; the source-side design was not tested (licence basis not recorded; Supplementary Table S2).

Prediction intervals

Without target labels, hierarchical conformal intervals²⁸ calibrated with regions as groups covered on average 0.86 of held-out observations (95% CI 0.80 to 0.92; nominal 0.90; four regions), and with four calibration regions they carry no finite-sample guarantee (internally pre-registered under an earlier protocol; Supplementary Table S10). Quantile intervals calibrated in Alaska by conformalized quantile regression²⁹ covered 0.32 to 0.73 in the four held-out regions (Supplementary Table S10). Across five ways of scaling the width of the hierarchical interval, coverage was 0.86 to 0.95 and mean width 81.3 to 132.4 cm (Fig. 4d). For the interval score³⁰, no difference was established between cell-specific flow scaling and a placebo with widths permuted within regions, or between any scaling and constant width. With ten labels, no difference in interval score was established between hierarchical predictive distributions and calibrated constant-width intervals (+9.5 cm; Holm-adjusted $P = 0.277$). With 40 and 160 labels, residual-ML intervals covered 0.84 to 0.90 in the Lena Delta, Canada and Alaska (Fig. 4e).

Discussion

The results define a label-count workflow for a new permafrost region (Fig. 6d). Without target labels, the reference is the source-coefficient (or year-matched) Stefan model, and ML enters only through physics pseudo-labels or a low-weight residual. With three to ten labels, the coefficient is recalibrated (lower error in one of four regions) and a low-weight residual added. From 40 labels, the method is selected by cross-validation within the target labels, with the gain arising in Canada. In this pool, the methods recommended from 3 to 160 labels had higher point-estimate error than the source-coefficient model [DECISION: Fig 7d *경로의 y 기준과 폴, 그림 명세 D-13*]. Within regions with thousands of labels, residual ML follows, with its gain between climate grid cells (Fig. 6a–c). Placement enters only as a pre-specified procedure, as spreading labels lowered error in Canada but not in the Lena Delta (Fig. 7).

The gains came from correcting the level of the Stefan coefficient E and bias at the climate-grid scale. E lumps soil thermal conductivity, the n-factor and the latent heat of soil water and ice³¹, which is why regional Stefan maps used land-cover-specific coefficients^{5,32}; here it also absorbs thawing-index bias and the label definition (Supplementary Note 1). With all labels, recalibration made up 85% of the four-region mean reduction by residual ML relative to the source coefficient, a mean dominated by W Russia (recalibration 13.5 of 14.0 cm; post hoc arithmetic). In Canada, whose coefficient was closest to the source value but varied between label and scoring blocks, ten-label recalibration increased error and the selection rule mostly chose a residual weight of 1.0, which we read as correcting label-location heterogeneity.

The zero-label results agree in direction with an Alaskan random-split comparison in which a random forest generalized less well than the Stefan model¹³. An earlier within-Alaska analysis [DECISION: *대회 보고서 인용 여부*] found a larger gain of a physics residual over the Stefan model, as the best of 56 settings, than the stricter test here (Supplementary Note 2). kNNDM²⁶ only mimics map-to-label distances in this clustered sample (13,606 Alaskan rows at 343 locations).

Alternative explanations are unlikely. Stefan pseudo-labels had lower error than shuffled and constant pseudo-labels (Fig. 3b), whereas a linear thawing-index placebo differed from them by less than 0.5 cm, so the gain reflects cell-level thawing-index information more than the square-root form. Recalibration gains held with single-year labels in W and E Russia and were 0.86 cm smaller with point labels than with 1 km location means in Canada (median, ten labels), which sets the label unit of the workflow. Physics and CatBoost results agreed between computing environments (maximum difference 3.6×10^{-14} cm) and a neural network differed by up to 5.6 cm, so the workflow uses only these deterministic methods.

The workflow does not rank sites for measurement or predict variation within climate grid cells; the 1 km maps (Supplementary Figs S1–S5) are a display resolution, and their information resolution is bounded by the climate (about 10 km) and soil (about 5 km) inputs. The conclusions are restricted to source data that include ground-penetrating radar labels, and climates warmer than the training range could not be evaluated. The Alaskan dominance of the source data (78–94% of source cells) matters most at small label counts, because the source coefficient carries the weight of ten labels in the recalibrated coefficient. Confirmatory evidence comes from four re-used

main regions and one new shallow region; the unblinded Tibetan Plateau is the only deep-regime region and is reported separately, so results are not generalized beyond these regions. Values depend on the test design (Fig. 1e), so randomly split studies are not directly comparable. Soil moisture, organic-layer thickness, snow duration and microtopography within grid cells are nearly absent from the covariates, so the within-grid share explained in Alaska (below 1%) describes these inputs, not a physical limit; on-site soil moisture did not explain the within-grid residual either (diagnostic; Supplementary Table S2). Method selection, placement, the label-rich tests, the bias correlation and the gain decomposition were designed after earlier results were viewed and are weaker evidence than the internally pre-registered contrasts.

Each step carries a condition: labels are 1 km location means, recalibration rests on the ten-label contrast, selection by cross-validation within the target labels depends on the number of 0.5° blocks covered by the labels and increased error relative to the fixed recipe in the Lena Delta with ten labels, and the whole workflow was tested only in new random splits of the same regions, with placement reduced to random sampling and non-inferiority holding only for the pooled mean, not in E Russia, Alaska or the Lena Delta. The data needed next are independent regions not used in development [PENDING: XF, new public regions; registered data deadline 11 October 2026, 14:22 KST; no eligible new macro region found so far], existing ALT maps scored on the same blocks^{15,33} (compared here with a registration deviation; Supplementary Table S2), and above all covariates that vary within climate grid cells, where 72% of the recalibrated Stefan squared error in Alaska lay. [PENDING: XE-a, XE-b, satellite inputs (stage 2); registered deadline 8 October 2026, 14:22 KST]

Methods

Study regions and labels

Active-layer thickness (ALT) labels of the main regions came from the GTN-P/CALM records³, the ALLena compilation for the Lena River Delta region³⁴ and version 2 of the ABoVE data set for Alaska and Canada³⁵. Records were aggregated to cells so that repeated measurements at one location do not enter as independent labels. A cell is a coordinate rounded to four decimal places (ABoVE and ALLena) or a CALM site, and its label is the mean of its observations (Supplementary Methods 1). The cell table has 17,572 rows, of which 17,467 carry a direct label from mechanical probing or ground-penetrating radar (GPR). Sixty-eight CALM cells whose metadata name ground temperature or thaw tubes are not used as labels. Twenty temperature-derived cells (15 in Alaska, 4 in Canada and 1 on the Tibetan Plateau) were found after the table had been frozen and remain in the direct class with a flag. Label years span 1990 to 2024, so predictions are static and refer to the 2015–2020 covariate climatology.

Macro-regions were defined geographically before the analysis, namely Alaska, the Lena Delta, Canada, W Russia, E Russia, Central Russia and Greenland [DECISION: 지역 이름 표기] [DECISION: 러시아 중부의 영문 표기]. Among Alaska and the four main regions, label rows range from 30 in E Russia to 13,606 in Alaska (Table 1). Ten sub-regions (Supplementary Table S11) were defined by k -means clustering of 0.5° block centroids without labels and were fixed before the main run.

The label count n is a row count, and the meaning of a row differs between regions. A row is the multi-year mean of a CALM site in W Russia and E Russia, usually a single-visit observation in the Lena Delta, mostly a point observation in Alaska and Canada, and a mean over about 1 km in the additional regions. Alaska, the Lena Delta and Canada have 39.7, 15.1 and 8.7 rows per 1 km location, and the 13,606 Alaskan rows lie at 343 locations in 74 blocks.

Additional regions were assembled from public archives under rules fixed before any fit (Supplementary Methods 2). A label had to be a direct end-of-season thaw depth from 1990 or later, at least 100 km from existing labelled regions, and was aggregated to a cell of about 1 km. A region was eligible for confidence intervals if the scoring blocks of its valid splits formed a union of at least eight blocks. Sources without a confirmed redistribution licence were removed in a licence-verified edition that defines the main verdicts (Supplementary Table S12). In this edition, Central Russia (57 rows, 13 blocks, shallow regime) and the Tibetan Plateau (132 rows, 34 blocks, deep regime) are eligible. A North Atlantic group (41 rows) is eligible only in the full edition and is reported in the Supplementary Information.

Covariates

All learning methods use 25 covariates that are available in every region. The six terrain covariates are elevation, slope, the sine and cosine of aspect, and the topographic position index and roughness in windows of 33×33 pixels, from the Copernicus DEM GLO-30 (Data availability). The eight climate covariates are mean annual air temperature, thawing degree-days (TDD), freezing degree-days, $\sqrt{\text{TDD}}$, warmest- and coldest-month air temperature, level-1 soil temperature and snow water equivalent, from ERA5-Land monthly means for 2015–2020²⁷. The nine soil covariates are clay, sand and silt content, bulk density, coarse fragments and pH at 5–15 cm and soil organic carbon at three depths, extracted from the 250 m SoilGrids 2.0 layers³⁶ resampled to about 5 km. The two model-based covariates are the 1997–2021 mean of the ESA CCI Permafrost ALT product, version 4.0, and its validity flag³⁷. This product is model output forced partly by the same reanalysis and is not treated as an observation.

TDD and the Stefan solution⁴ are

$$\text{TDD} = \sum_m \max(\bar{T}_m, 0) d_m, \quad \text{ALT} \approx E s, \quad s = \sqrt{\text{TDD}}, \quad (1)$$

where $\bar{T}_m$ is the monthly mean 2 m air temperature of month m , d_m is its number of days and E is in $\text{cm} (\text{°C d})^{-1/2}$. A cell is scored only if its label, its CCI value and its TDD from level-1 soil temperature are valid, and all methods share the scoring cells. Quantities derived from labels, such as the count and years of observations, are excluded from the inputs, and unit tests enforce this. The information resolution of the inputs is about 0.1° for climate and about 5 km for soil.

Stefan baselines and coefficient recalibration

The coefficient E absorbs soil thermal properties, water content and surface conditions, and regional applications calibrate it against local measurements⁵. The source set S of a target contains all labelled cells outside the target region and a 100 km buffer around it, and the source coefficient is the least-squares slope through the origin,

$$E_0 = \frac{\sum_{i \in S} s_i y_i}{\sum_{i \in S} s_i^2}, \quad (2)$$

where y_i is the observed ALT. The source-coefficient Stefan model predicts $E_0 s$. Alaskan cells form 78–94% of the source sets of the main targets, so E_0 varies little between these targets (1.59–1.62 cm (°C d)^{−1/2}; Table 1). With E_{ls} the same estimator computed from the n target labels, the recalibrated Stefan model predicts $E_n s$ with

$$E_n = \frac{n E_{ls} + \kappa E_0}{n + \kappa}, \quad \kappa = 10. \quad (3)$$

The source coefficient thus carries the weight of ten labels, because recalibration with few labels can increase error²⁵. The value of κ was fixed from earlier experiments of this project and is not blind, and $\kappa = 3$ and $\kappa = 30$ are sensitivity cases. The local least-squares Stefan model uses E_{ls} without shrinkage. A gain of machine learning (ML) is stated only against the recalibrated model fitted to the same labels, because the gain over the source coefficient includes the gain of recalibration.

A registered comparison of physics models and ALT products added two stronger baselines. The year-matched Stefan model applies E_0 to $\sqrt{\text{TDD}}$ averaged over the observation years of each label within 2010–2024 (2010–2014 for earlier labels) and is reported at zero and at all labels. At 40 and 160 labels, the output of a soil-property Stefan model (Supplementary Methods 3), recalibrated with equations (2) and (3), is reported as the stronger recalibrated baseline.

Machine-learning methods and combination structures

In the transfer design, training rows are the source cells and the n target labels with equal weight. Anchored methods have the form

$$\hat{y}(x) = a(x) + \lambda g(x), \quad (4)$$

where x denotes the covariates of a cell, a is a physics anchor and g is a learner trained on residuals from it. Direct ML regresses ALT on the covariates, which include $\sqrt{\text{TDD}}$ and the CCI value. Physics pseudo-label augmentation adds rows with value $E_n s$ at cells of the label half of the target, ten per source row, while drawn cells keep their observed value. Source anchor plus residual ML uses $a = E_0 s$. Recalibrated anchor plus residual ML (residual ML) uses $a = E_n s$ and residual targets $y - E_0 s$ for source rows and $y - E_n s$ for target rows. Augmentation plus anchor plus residual adds zero-valued residual rows at label-half cells. Physics-input ML adds the outputs of a Kudryavtsev-type model³⁸ and of the soil-property Stefan model, or the anchor

value, to the inputs of direct ML. The Stefan-CCI anchor is the mean of $E_n s$ and the CCI value where that value is valid. Four further methods of the main experiment are described in Supplementary Methods 3. The residual weight $\lambda \in \{0.25, 0.5, 1.0\}$ is applied after fitting. The reference weight for verdicts, fixed before the main run, is 0.25 for residual structures and 1.0 for direct and physics-input structures, and the source anchor plus residual at 0.25 is called the low-weight residual.

Four placebos tested whether the gain of augmentation comes from the cell-level Stefan values. They replaced the pseudo-labels by the same values shuffled among cells, by their target-level mean, by the mean source ALT, or by a function linear in TDD fitted to the source.

The learner of the method comparisons is CatBoost³⁹ (200 iterations, learning rate 0.05, depth 3, two seeds). For direct ML without target labels, ten learners were compared. They were this setting, a larger CatBoost, CatBoost tuned by leave-one-source-region-out cross-validation, a random forest, TabPFN v2⁴⁰ and TabICL v2⁴¹ without fine-tuning, and four neural networks with hyperparameters selected by leave-one-source-region-out cross-validation on source data^{42,43}. The networks are a multilayer perceptron, a multi-head perceptron (not TabM⁴⁴), a reduced FT-Transformer-type model⁴⁵ and RealMLP⁴⁶ (settings in Supplementary Methods 5). In the figures these methods are labelled Source Stefan, Year-matched Stefan, Recalibrated Stefan, Anchor + residual ML, Source anchor + residual, Physics pseudo-labels, Direct ML, Physics-input ML, Stefan-CCI anchor and, for the rule described below, Target-label CV selection.

Transfer design

To prevent information leakage from nearby labelled cells, the source of a target excludes the target region and a 100 km buffer. Sub-regions were evaluated with the parent region excluded from the source and, as a simulated new area inside a labelled region, with the rest of the parent kept. The 27 targets are the six version-3 regions other than Alaska, Alaska and the ten sub-regions in both modes.

The 0.5° blocks of each target were split five times into halves A and B by assigning randomly permuted blocks to the half with fewer cells, so that labelled and scored cells are separated by block boundaries¹⁶. Labels were drawn from A and errors were scored on the masked cells of B. For each n in $\{0, 3, 10, 40, 160, 320, 1000\}$ below the size of A, and for all cells of A, five random draws were made with seeds fixed by target, mode, split, n and draw, so that all methods used the same labels. W Russia and E Russia have 14–17 cells in A, so pooled means at 40 and 160 labels contain only the Lena Delta and Canada.

The four main regions (the Lena Delta, Canada, W Russia and E Russia) form the confirmatory pool. They were used in earlier experiments of this project, so their verdicts are re-tests of re-used regions. Alaska is a reference target, reported separately and in an auxiliary three-region mean with the Lena Delta and Canada. Central Russia and Greenland of the version-3 table are reported as point estimates, and sub-regions, which share cells with their parent, are counted by verdict class. The additional regions enter extended pools of five shallow regions and of six regions including the Tibetan

Plateau. All Tibetan scored cells lie above the 99.5th percentile of source-cell elevation, so Tibetan contrasts are not used as evidence for independent regions.

A validation ladder scored direct ML and a Stefan model fitted to the training folds on Alaska and the four main regions under eight schemes of increasing designed separation, from random cells to region holdout (Supplementary Methods 3). An earlier Alaskan analysis added k -fold nearest-neighbour distance matching (kNNDM)²⁶, which matches test-to-training distances to those between a prediction grid and the labels.

Within-region design and gain decomposition

Within-region tests ask what ML adds where hundreds to thousands of labels exist, and they were designed after the transfer results had been viewed. In Alaska, the Lena Delta and Canada, labels were drawn from half A and were the only training rows. The source coefficient of the transfer design served as prior, λ was chosen from $\{0.25, 0.5, 1.0\}$ by block cross-validation within the drawn labels, and augmentation used ten pseudo-labels per label. A first test (5 splits, 1000 or more labels, comparator the best cross-validated physics calibration, nine SAR covariates added in Alaska) was rejected. A second test changed four elements, namely the comparator (the recalibrated Stefan model), the splits (25, of which 24 were valid in the Lena Delta), the recipes (Stefan-CCI anchor and direct ML added) and the label range (200, 500, 1000 and all labels). Both tests are reported.

The gain decomposition asks at which spatial scale ML reduces error. Scoring cells were grouped by 0.5° block and air $\sqrt{\text{TDD}}$, so that Stefan predictions and the eight climate covariates are constant within a group. With N_g cells and means $\bar{y}_g$ and $\bar{\hat{y}}_g$ in group g ,

$$\text{SSE} = \sum_g \sum_{i \in g} [(y_i - \bar{y}_g) - (\hat{y}_i - \bar{\hat{y}}_g)]^2 + \sum_g N_g (\bar{y}_g - \bar{\hat{y}}_g)^2, \quad (5)$$

where the terms are the within-grid and between-grid components. The within-grid RMSE uses groups with at least two cells, and for a Stefan model it equals the within-group standard deviation of the observations. Shares of the reduction are computed from the SSE because RMSE differences of components are not additive. The registration named soil $\sqrt{\text{TDD}}$ for the grouping, whereas the run used the air $\sqrt{\text{TDD}}$ of the Stefan predictions. A soil-grouping sensitivity analysis was therefore registered and run (adjusted Rand index between groupings 0.999 in Alaska, 0.997 in Canada and 0.832 in the Lena Delta).

Label placement procedures

Placement strategies select cells of half A without using label values, except for active selection. Besides cell-random sampling, they were block stratification, covariate-cluster stratification, covariate max-min distance sampling, source-extrapolation priority, active selection by the prediction variance of the residual learner and $\sqrt{\text{TDD}}$ priority (Supplementary Methods 8). They were compared within regions at 20 to 500

labels and, for block and max-min sampling, in transfer at 10 and 40 labels. These tests were designed after results had been viewed, whereas a block-spread contrast of the main experiment was registered before results.

Algorithm P combines block allocation with covariate coverage and is the pre-specified placement procedure tested in the workflow analysis. It uses the standardized covariates of candidate cells (no label values), block sizes N_b , a nested budget of 10, 40 and 160 labels, an allocation exponent γ and a seed.

1. Order blocks by farthest-first traversal of their covariate centroids μ_b , starting from a seeded block or the block of existing labels.
2. At budget n_k , allocate $q_b = n_k N_b^\gamma / \sum_{b'} N_{b'}^\gamma$ labels to block b by largest remainders, capped at N_b .
3. Fill the block with the largest remaining deficit, with its cell nearest to μ_b if it has no label and otherwise with its cell farthest from all chosen labels.
4. Keep earlier labels when the budget grows.

The candidate settings are $\gamma = 0, 0.5$ and 1 , a variant with farthest-first choice of the first cell in a block, block stratification and max-min sampling. One candidate is selected by a mechanical rule from Alaskan targets only, so that the evaluation regions do not inform the choice. Under this rule all candidates were excluded, so Algorithm P is cell-random sampling, and the workflow contrasts of Algorithm P against random placement were not tested. The placement step of the workflow therefore contributed nothing; with ten labels the workflow and its random-placement comparator use the same labels and model, and with 40 and 160 labels the comparison with random placement and the fixed residual recipe compares the selection rule with the fixed recipe on the same labels. Following the registered reporting rule, no map that ranks measurement locations was produced.

Method-selection rule and bias diagnosis

The selection rule chooses a method from the target labels alone.

1. With fewer than ten labels, use the recalibrated Stefan model.
2. Assign the blocks of the labels at random to at most five folds (cell folds if one block). With fewer than two folds, use the recalibrated Stefan model.
3. In each fold, recompute E_n and fit seven candidates, namely the source-coefficient, recalibrated and local least-squares Stefan models, residual ML with $\lambda = 0.25$ and 1.0 , augmentation plus anchor plus residual and physics pseudo-labels.
4. Choose the candidate with the lowest cross-validated RMSE (ties to the smaller λ and the earlier candidate).
5. Refit it on all labels.

The rule was registered after the label-grid results had been viewed and was applied at 10, 40, 160 and all labels to 30 targets (the 27 transfer targets and three licence-verified additional-region targets). It was compared with a fixed residual recipe with a non-inferiority margin of 0.5 cm.

The bias diagnostic is the absolute mean error of the source-coefficient Stefan model at the ten labels of a draw, averaged over draws and splits. It uses only labels

that recalibration also uses, so it is known before any learner is fitted. Its Spearman correlation with the gain of the selection rule over the source-coefficient model at all labels was computed over 28 targets. This gain includes recalibration, so the correlation concerns label-based correction (recalibration plus residual). The part of the gain beyond recalibration was examined in a separate post hoc analysis (Additional registered analyses).

Statistical analysis

With SSE_b and N_b the sum of squared errors and the number of scoring cells in block b of half B , two root-mean-square errors (RMSE, cm) were computed,

$$\text{RMSE}_{\text{cell}} = \left(\frac{\sum_b SSE_b}{\sum_b N_b} \right)^{1/2}, \quad \text{RMSE}_{\text{block}} = \frac{1}{|B|} \sum_b \left(\frac{SSE_b}{N_b} \right)^{1/2}, \quad (6)$$

where $|B|$ is the number of scoring blocks. The cell-weighted value is primary, and the block-equal value limits the influence of densely sampled blocks. A contrast Δ is the RMSE of the first method minus that of the second, paired by split, draw and seed and averaged over splits, so that negative values mean lower error for the first method. Stratified means weight regions equally.

Confidence intervals (CIs) are 95% percentile intervals of a bootstrap of the scoring blocks within each split, with the same resampled blocks for both methods. The hypotheses aggregated by the label-grid harness (the main experiment and the additional-region runs) and the prediction-interval experiment used 1000 replicates, and all other contrasts used 10,000. The registered block-spread contrast of the main experiment and the workflow analysis, whose arms use different label sets, used a two-stage bootstrap that also resamples draws; the placement tests report intervals conditional on the label draws. An auxiliary interval resampled the union of a region's blocks once for all splits. Where it changes the verdict, the result is described as depending on the split-independence assumption, and statements without this qualification require both verdicts to agree. A region entered a pooled interval if its valid splits had at least eight scoring blocks in their union.

A contrast is lower error if the upper limits of both the cell-weighted and block-equal CIs are below zero, higher error if both lower limits are above zero, equivalent if neither holds and all four limits lie within ± 0.5 cm (two one-sided tests at 2.5%⁴⁷), and undetermined otherwise ("no difference was established"). The margin was fixed before any result. It is about 1–2% of the cell-weighted RMSE of the source-coefficient Stefan model in the main regions (21.7–43.0 cm) and is a convention, not a measurement error. Margins of 1.0 cm and of 2% of that RMSE are sensitivity cases. A lower- or higher-error contrast with $|\Delta| < 0.5$ cm is reported as distinguishable with a magnitude below 0.5 cm. Non-inferiority is one-sided and requires both upper limits to be below +0.5 cm. The minimum label count n^* of a contrast is the smallest n at which the upper limit of the cell-weighted CI is below zero and at least two thirds of splits and 75% of draws have $\Delta < 0$.

The significance level is $\alpha = 0.05$ two-sided, corresponding to 0.025 one-sided. A P value is the larger of the two-sided bootstrap P values of the two weightings,

and unadjusted and Holm-adjusted P values of all registered contrasts are listed in Supplementary Table S1. Ten contrasts fixed before the results are the only ones that may carry verdict words in the abstract. Their P values were adjusted by Holm’s procedure⁴⁸ with the family size fixed at ten, and a direction is stated in the abstract only if the adjusted P is below 0.05. Otherwise the abstract states that no difference was established and the main text adds “significant before correction only”. Results of the within-region, placement and selection tests and of the additional analyses appear in the abstract as numbers only. Within other registered families, Holm-adjusted P values are auxiliary, except in the workflow test and the product comparison, where they set the wording.

Pooled contrasts rest on four to seven regions, so the resampling unit is the 0.5° block (13 to 74 per region), and region values and ranges accompany every pooled mean. For statements about regions in general, key contrasts were recomputed over 50 splits in the four main regions and Alaska and summarized by a sign test, a sign-flip permutation test and a Hartung-Knapp random-effects mean⁴⁹. Such statements are made only if the Hartung-Knapp interval excludes zero, and with four regions the smallest two-sided sign-test P is 0.125. Rank correlations are Spearman coefficients with intervals from joint resampling of blocks and draws. An error floor, the pooled variance of observations within groups of at least five cells sharing coarse non-terrain covariates (13,333 cells in Alaska), serves only as a denominator of effect sizes.

Analyses designed after earlier results had been viewed are labelled as such, and post hoc descriptions carry no verdict words. Each registered hypothesis carries one of four blinding labels. These are blind, partly unblinded (direction seen under another protocol), replication (unblinded) (direction seen for the same contrast) and results existed at registration (unopened). Prediction intervals were evaluated by empirical coverage and by the interval score $(U - L) + (2/a)(L - y)_+ + (2/a)(y - U)_+$ of a central interval $[L, U]$ with $a = 0.1$ ³⁰. Hierarchical conformal intervals used source regions as groups²⁸ and, with two to four groups, carry no finite-sample guarantee (Supplementary Methods 4). No values are given after $\pm$.

Internal pre-registration and deviations

Hypotheses, contrasts, decision rules and permitted sentences were committed to the project’s git repository before the results they govern were retrieved (internal pre-registration). The commits were not time-stamped by a third party, and times are committer times in Korea Standard Time (full table in Supplementary Methods 6). The label-grid experiment (LG, L1–L8) was registered in commit 40be64c on 29 September 2026 at 15:45:46. Its extension (LGX), the foundation-model and additional-region runs (LGT, LGD), the interval experiment (LGU), the TabICL and tuned-network runs (LGF) and the reporting rules with the abstract bundle AB1–AB10 (WRAPUP, 316714c, 30 September 2026, 03:02:42) followed, all before the first retrieval of results at 07:20:22 that day.

The workflow experiments were designed after these results had been viewed and were registered before running, WF0–WF5 in 4168331 (1 October 2026), WF6–WF8 in 1b42b4d (1 October 2026), WF9 and WF10 in 8180632 (2 October 2026) and the soil-grouping sensitivity in 7da0064 (2 October 2026). WF0 is a post hoc description,

and the priority map planned as WF5 was not produced. The analyses XA to XJ were registered in commit 2678100 (4 October 2026, 14:22:39) before any fit. Only XC-F3 is labelled a confirmatory test registered before running, and XA is a post hoc analysis. The placement of each analysis in the main text or the Supplementary Information was fixed before its results.

Deviations and post hoc changes are as follows.

1. The gain decomposition (WF9) grouped cells by air rather than the registered soil $\sqrt{\text{TDD}}$, and a soil-grouping sensitivity analysis was added.
2. After the first within-region test (WF1-a) was rejected, the second test (WF6) changed comparator, splits, recipes and label range.
3. The product comparison (XG) was registered as a descriptive table conditional on an interval hypothesis (LGU-B2). After that hypothesis was undetermined, XG was re-registered with six Holm-adjusted contrasts and carries a registration-deviation label.
4. The workflow splits (XC) moved from seeds 6–15 to 201–210 because seeds 1–200 had been used (LGX-N4), and a proposal to use XC verdict words in the abstract was withdrawn.
5. AB4 carries a note that a pre-result sensitivity calculation had displayed the split distribution of the same quantity.
6. Twenty temperature-derived cells found after the table had been frozen remain in the direct class with a flag.
7. Four technical amendments (ff1be02, 30 September 2026, 11:44:30) were committed after the first retrieval but before any result was unpacked or opened, and they change no hypothesis, contrast or verdict rule. [DECISION: 수용 여부, 방법 초안 183 행]
8. CatBoost did not pass cross-environment check (i), so extended-pool rows that combine platforms carry a cross-environment flag.
9. No map of measurement priorities was produced, following the registered rule.
10. The climate-extrapolation test (WF10) was not decidable because its extrapolation set had three blocks in Canada.
11. XA resampled eight target families instead of the seven registered, because the North Atlantic target of its 28-target set was not in the registered family list (implementation deviation recorded with the sealed results).

Additional registered analyses

The ten analyses XA to XJ follow the conventions above, and their placement in the main text or the Supplementary Information was fixed before results (Supplementary Methods 9; results in Supplementary Table S2).

XA (main text, post hoc) splits the gain over the source-coefficient model into the shares of recalibration, of remaining shrinkage and of ML beyond recalibration, and estimates their rank correlations with coefficient error under two definitions, with intervals from joint resampling of target families and blocks. The registered list had seven families; the North Atlantic target, which was not in that list, was resampled as an eighth single-target family, and the 27 targets of the seven registered families were

analysed as an unregistered sensitivity with the same result categories. Results are stated only as an interval excluding zero or as not established. XB (Supplementary) learns convex weights of physics models and ALT products from the target labels by cross-fitting, with and without a residual learner. XC (main text) applies Algorithm P, the ten-label diagnostic and the method rule in sequence on new splits of the five shallow regions. Its eight Holm-adjusted hypotheses compare the workflow with random placement and a fixed residual recipe at 40 and 160 labels, test non-inferiority to the recalibrated Stefan model at 10, 40 and 160 labels, compare Algorithm P with random placement under the selection rule, and test the workflow at ten labels in newly obtained regions. The evaluation cells were used in designing the components, so all but the last hypothesis are partly unblinded.

XD (Supplementary) selects the setting of Algorithm P on Alaskan targets, tests alternative coefficient estimators and evaluates a learned placement policy once outside Alaska without P values. XE (Supplementary) adds public inputs at 10–500 m to residual ML within regions and compares within-grid RMSE with the 25-covariate model and the recalibrated Stefan model. XF (Supplementary, blind) applies the eligibility rules to new public data and repeats core contrasts in each eligible region. XG (main text) scores existing ALT maps on the same blocks after masking blocks within 5 or 25 km of product training sites, raw against the source-coefficient model and recalibrated against residual ML. XH (main text) describes, without verdict words, the errors of the recalibrated Stefan model, residual ML and direct ML under random-cell, site, block, kNNDM and region-holdout validation in three regions. XI repeats the climate-extrapolation test on the licence-verified expanded Canadian table, and XJ tests temperature-derived labels as low-weight auxiliary rows (both Supplementary).

Computing environments and reproducibility

Physics models and CatBoost run deterministically on CPUs. The label-grid experiment and its extension ran on cloud nodes (Rescale) with 64 CPU cores and four NVIDIA T4 GPUs, and the workflow experiments and additional analyses on cloud CPU nodes of one processor family (AMD EPYC Milan, 96 cores, and Milan-X, 64 cores). The foundation models, tuned networks, interval experiment and additional-region runs used a shared local server with NVIDIA RTX 3090 GPUs. Package versions were pinned (CatBoost 1.2.10 on all platforms; Supplementary Table S13), and each output records code hash, configuration hash, package versions and device.

Physics and CatBoost results agreed between cloud and local runs to within 3.6×10^{-14} cm over 34,360 keys, and the two cloud node types gave identical errors over 11,814 common keys. Neural networks did not reproduce across platforms. Of 288 FT-Transformer keys, 43 differed by more than 0.5 cm (maximum 5.61 cm) although refits within a platform were identical, and CatBoost on a ridge anchor differed by up to 0.46 cm. Both sides of a contrast therefore come from one platform, pooled means across platforms carry a cross-environment flag, and the workflow uses only CatBoost and closed-form physics.

Maps and colour scales

Maps use a north polar stereographic projection with true scale at 70° N and the central meridian stated in each legend, and the Tibetan Plateau appears in a Lambert azimuthal equal-area inset. Land is from Natural Earth (1:50m), permafrost zones from the ESA CCI Permafrost extent product, version 4.0 (permafrost fraction, 1997–2021 mean), and maps were drawn with Cartopy 0.23.0. Scientific colour maps^{50,51} were used, oslo reversed and restricted to 0.12–0.90 for ALT, broc centred on zero for error differences, bam for residuals and acton reversed and restricted to 0.05–0.95 for interval width.

Use of large language models

[DECISION: LLM 사용 기재 문구]

Code availability

Code for data preparation, experiments, statistics, figures and tests is available at [DECISION: 공개 저장소 주소와 공개 시점] and archived with a version DOI [DECISION: Zenodo DOI]. The main components are the label-grid experiment (`scripts/3_deep_learning/h40_label_grid.py`), the validation ladder (`scripts/2_evaluation/h41_validation_ladder.py`), the extension experiment (`scripts/3_deep_learning/h42_label_grid_ext.py`), the workflow experiments (`scripts/3_deep_learning/h54_workflow.py`), the abstract-contrast aggregator (`scripts/2_evaluation/h39_scenarios.py`), the post hoc reanalysis (`scripts/2_evaluation/wf0_reanalysis.py`), the additional analyses (`x_*.py`), the unit tests (`tests/`) and the cloud job configurations (`configs/rescale/`), together with the registration documents and their revision histories. The additional analyses pin catboost 1.2.10, scikit-learn 1.9.1, pandas 2.3.3, scipy 1.17.1 and numpy 2.1.1. The pretrained weights of TabPFN v2 and TabICL v2 are not redistributed and are obtained from their providers.

Data availability

The active-layer labels of the main regions are available from the GTN-P/CALM compilation³ (PANGAEA, <https://doi.org/10.1594/PANGAEA.972777>), the ALLena compilation for the Lena River Delta³⁴ (PANGAEA, <https://doi.org/10.1594/PANGAEA.973813>) and the ABoVE soil moisture and active-layer thickness data set, version 2³⁵ (ORNL DAAC, <https://doi.org/10.3334/ORNLDAAAC/2369>). The labels of the additional regions come from published data sets^{52–65}, and the temperature-derived labels used only in the label-definition test come from a Tibetan Plateau compilation⁶⁶. Their identifiers and licences are listed in Supplementary Table S12. Labels from sources without a confirmed redistribution licence were excluded from the licence-verified edition used for the main results and are not redistributed; they can be obtained from the original providers listed in Supplementary Table S12. The full North Atlantic edition, which includes such labels, is reported in the Supplementary Information only.

The covariates were derived from ERA5-Land monthly averaged data²⁷ (Copernicus Climate Data Store, <https://doi.org/10.24381/cds.68d2bb30>), SoilGrids 2.0³⁶ [미확인: 접근 시점의 SoilGrids 판 표지], the Copernicus DEM GLO-30 (Copernicus DEM, Global and European Digital Elevation Model, <https://doi.org/10.5270/ESA-c5d3d65>; tiles from the AWS open-data bucket copernicus-dem-30m) and the ESA CCI Permafrost active-layer thickness product, version 4.0³⁷ (CEDA, <https://doi.org/10.5285/d34330ce3f604e368c06d76de1987ce5>). The synthetic-aperture radar (SAR) covariates of the within-Alaska test came from the ABoVE airborne L- and P-band SAR active-layer retrievals for 2017 (ORNL DAAC, <https://doi.org/10.3334/ORNLDAAC/2004>) and from upscaled P-band polarimetric SAR active-layer maps of northern Alaska⁶⁷ [미확인: PolSAR 상향 ALT 자료의 DOI, 후보 10.3334/ORNL-DAAC/2332]. The map masks use the ESA CCI Permafrost extent product, version 4.0 (permafrost fraction, annual files 1997–2021) [미확인: 영구동토 범위 자료 v4.0의 DOI] and the JRC Global Surface Water occurrence layer⁶⁸.

The assembled label and covariate tables, the run outputs, the aggregate and verdict tables, the cell-level predictions and the source data of all figures will be deposited at [DECISION: 저장소와 DOI]. Rows from sources without a confirmed licence are excluded from the deposited tables. [DECISION: KPDC 자료, SI 포함 여부와 식별자, 이용 조건]

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Figure legends

Figure 1. Study regions, Stefan coefficients and evaluation design. (a) Labelled cells of the licence-verified label tables in 0.5° blocks; circle area is proportional to the number of 1 km label locations in a block, colour marks the region as in b (grey, Greenland, Svalbard and Scandinavia), and New regions marks the regions added after the label tables were fixed (Central Russia, expanded edition; Tibetan Plateau, inset; Table 1). Shading shows continuous (permafrost fraction $\geq 90\%$) and discontinuous (50–90%) permafrost (ESA CCI Permafrost fraction v4.0, 1997–2021 mean); rectangle, area of c and d. (b) Stefan coefficient $E = \text{ALT}/\sqrt{\text{TDD}}$ of labelled cells by region (logarithmic axis), where ALT is the active-layer thickness and TDD the ERA5-Land²⁷ air thawing index ($^\circ\text{C d}$, 2015–2020 mean); up to 400 cells per region, median and interquartile range; Source coefficient, the coefficient fitted on the source pool when that region is held out. (c) Region holdout: labelled cells of the Lena Delta and all cells within 100 km (Buffer) are removed from the training data. (d) Split of the same region into label and scoring blocks (one of five splits), with 40 label cells drawn from the label blocks and errors scored only in the scoring blocks. The within-region tests use the same block split without source data and were designed after the transfer results had been inspected. (e) RMSE against median distance from scoring to nearest training cell, from random-cell to region-holdout validation (lines, three-region means; symbols, regions with cell-weighted 95% intervals; open, Stefan with source coefficient). From random-cell to kNNDM splits, direct-ML RMSE increased by 5.76, 8.17 and 14.20 cm and recalibrated-Stefan RMSE by 0.29, 0.51 and 4.28 cm (Alaska, Lena Delta, Canada; cell-weighted; descriptive, designed after earlier results were viewed). Cells without a verified licence (Canada 38, North Atlantic 19, W Russia 1) are not drawn. Polar stereographic projection (central meridian 127° E ; scale bar true at 70° N), also in c and d; Tibetan Plateau inset, Lambert azimuthal equal-area projection with its own scale bar. Coastlines, Natural Earth 50 m (10 m in c and d); Cartopy 0.23.

Figure 2. With ten labels, Stefan recalibration lowered the four-region mean error; no further difference was established for residual ML. Error change is the root-mean-square error (RMSE) of a method minus that of the source-coefficient Stefan model (zero line) on the scoring blocks of each held-out region; negative values mean lower error. Recalibrated Stefan refits the coefficient E with the n target labels, shrunk towards the source coefficient ($\kappa = 10$); Anchor + residual ML adds a CatBoost residual with weight 0.25; Direct ML is CatBoost without a physics anchor; Year-matched Stefan uses thawing degree-days of the label years. (a) Mean of the four main regions (the Lena Delta, Canada, W Russia, E Russia), up to ten labels. (b) Mean of the Lena Delta and Canada, the regions with at least 40 candidate labels; All, every label; Year-matched Stefan is a point estimate. (c to f) Single regions; label counts above the candidate pool (14–17 cells in W and E Russia, 325–406 in Canada) are masked, and panel e has its own vertical range (values to -12.5 cm). Lines are cell-weighted means over five splits; bands are 95% block-bootstrap confidence intervals (CIs) of the two recalibration-based methods (10,000 resamples for the fixed-composition means in panels a and b; 1000 in panels c to f). With ten labels, recalibration lowered the four-region RMSE by 2.5 cm (95% CI 1.9 to 3.0 cm; Holm-adjusted $P = 0.001$), with lower

error in W Russia only (12.0 cm; 95% CI 10.5 to 13.6 cm) and higher error in Canada (2.3 cm; 95% CI 0.7 to 2.9 cm); adding the residual changed it by -0.18 cm (95% CI -0.53 to -0.01 cm; $P = 0.034$ before correction, Holm-adjusted $P = 0.136$). Verdicts of all plotted contrasts are in Supplementary Table S1. The label-grid analyses were internally pre-registered and run on one computing platform.

Figure 3. Physics pseudo-labels lowered error relative to shuffled pseudo-labels, and the better combination structure depended on the number of labels. (a) Concept on the 31 labelled cells of W Russia: active-layer thickness against the square root of thawing degree-days, with the source-coefficient Stefan line (Source Stefan, $1.59 \text{ cm } (^{\circ}\text{C d})^{-1/2}$), the line recalibrated with ten drawn labels (Recalibrated Stefan, 2.11) and the departures of those labels that residual ML learns (Residual); thickness increases downwards. (b) Direct ML trained with physics pseudo-labels minus the same model trained with placebo pseudo-labels, four-region mean, without labels and with ten labels. (c) Residual structure minus physics-input structure against the number of labels, for two input sets (Physics inputs, outputs of two physics models; Recalibrated inputs, source and recalibrated Stefan predictions); the key separates registered label counts (0 and 10) from auxiliary ones, and values at 40 and 160 labels are Lena Delta and Canada means. (d) Local least-squares minus shrinkage recalibration of the Stefan coefficient (descriptive; positive values favour shrinkage), four-region mean up to ten labels, Lena Delta and Canada mean beyond. Intervals are 95% block-bootstrap confidence intervals (CIs; 10,000 resamples of 0.5° scoring blocks), weighted by cell and by block as indicated in the key (panel d, by cell); the grey band marks ± 0.5 cm. In panel b, physics pseudo-labels lowered error relative to shuffled pseudo-labels by 1.6 cm without labels (95% CI 1.0 to 2.0 cm; Holm-adjusted $P = 0.001$); the difference from linear thaw-index pseudo-labels was 0.48 cm without labels and within ± 0.5 cm with ten labels. In panel c, the residual structure had 2.5–3.7 cm lower error under both weightings with up to ten labels (Holm-adjusted $P = 0.001$ for physics inputs at ten labels); at 40 and 160 labels the physics-input structure had 1.9–2.4 cm lower error, a difference that came from Canada (one of four contrasts depends on the split-independence assumption). The placebo and structure analyses were internally pre-registered.

Figure 4. Without target labels, 2 of 30 targets lost over 2 cm with physics pseudo-labels and 18 with direct ML. (a, b) Error change relative to the source-coefficient Stefan model for direct ML (a) and direct ML with physics pseudo-labels (b), at each target’s labelled-cell centre (17 targets; source excludes the parent region). Values below -40 cm take the end colour (Tibetan Plateau in a, -60.0 cm). Alaska targets enlarged; Lena Delta targets offset with leader lines. Title counts are post hoc over 30 targets (adding within-region sub-regions and three expanded-label editions; point-estimate threshold). (c) Direct ML by learner and two physics baselines, four-region mean. The random forest, CatBoost with physics-model inputs, and four neural networks configured by leave-one-source-region-out cross-validation had higher error (2.4–5.2 cm). The other five learners, including the main CatBoost learner (2.2 cm; 95% CI 1.1 to 3.4 cm; Holm-adjusted $P = 0.023$), had higher error under the primary intervals, dependent on the split-independence assumption. Year-matched Stefan had 1.0 cm lower error, also dependent on that assumption. Anchor ensemble, mean of

scale-calibrated Stefan, Kudryavtsev and ESA CCI anchors. Existing ALT maps (registration deviation; designed after earlier results were viewed; 5 km training-site blocks removed): CCI v5 had higher error (+23.79 cm at axis end; 95% CI 15.34 to 30.16); for Wei v2 (Lena Delta and Canada; 47.4% imputed) no difference was established. (d) Coverage of 90% prediction intervals against mean width without labels for five interval scalings: four-region mean (95% bootstrap intervals, 1000 resamples) and regional values; stacked labels, two near-coincident means (upper, higher coverage); line, nominal 0.90. (e) Coverage of residual-model intervals with 40 and 160 labels against width relative to zero labels; band, 0.85 to 0.95. Intervals in c and e: 95% block-bootstrap confidence intervals (10,000 resamples of 0.5° scoring blocks), in c weighted by cell and block (key); ± 0.5 cm band, equivalence margin. Polar stereographic projections (central meridian 127° E; Alaska enlargement, 157° W), scale bars true at 70° N; Tibetan Plateau inset, Lambert azimuthal equal-area projection. Permafrost zones, ESA CCI Permafrost fraction v4.0 (1997–2021 mean); coastlines, Natural Earth 50 m; Cartopy 0.23.0.

Figure 5. The error reduction of label-based correction was rank-correlated with the source-coefficient bias measured with ten labels. (a) Smallest tested number of labels n^* (interval from the previous grid value) at which recalibration lowered error relative to the source-coefficient Stefan model (Recalibration) and residual ML lowered error relative to the recalibrated model (Residual ML), under three registered conditions (Methods); Not reached marks targets without n^* up to the largest tested count. Rows follow the ten-label bias of b; sub-regions are not independent. (b) Error change of the method chosen by cross-validation within the target labels, relative to the source-coefficient Stefan model with all labels, against the absolute mean bias of that model at the first ten labels (Spearman $\rho = 0.38$; 95% CI 0.17 to 0.55; 28 targets; signed bias, $\rho = -0.01$). The Tibetan Plateau lies outside the axes (bias 227.7 cm; error change -159.6 cm in b, -135.7 and -5.9 cm in d); the North Atlantic is a point estimate. (c) Cross-validation selection minus a fixed residual recipe ($\lambda = 0.25$). The selection was non-inferior (margin 0.5 cm) at 40 and 160 labels in the mean of the Lena Delta and Canada, with 1.3 cm lower error at 160 labels; the 0.87 cm reduction at 40 labels and non-inferiority with all labels depend on the split-independence assumption, and non-inferiority did not hold at ten labels. The gain came from Canada (2.1 and 2.7 cm); the Lena Delta changed by $+0.35$ and $+0.07$ cm. (d) Error change from recalibration and from residual ML beyond the better recalibrated Stefan model with all labels, against the same bias. Relations of the bias with the error reduction from recalibration ($\rho = 0.37$; 95% CI -0.11 to 0.93) and beyond recalibration ($\rho = 0.28$; -0.19 to 0.80 ; 0.18 to 0.54 with targets fixed) were not established (post hoc analysis). Intervals are 95% confidence intervals (CIs) from 10,000 resamples of 0.5° scoring blocks (plus target families in d), weighted as in the key; the grey band marks ± 0.5 cm. Analyses in b and c were designed after earlier results had been inspected.

Figure 6. Within Alaska, residual ML gains over recalibrated Stefan were below 1 cm and came from climate-grid-scale bias correction. (a) Within-region tests (half of the 0.5° blocks scored, 25 splits): error change of Anchor + residual ML (cross-validated residual weight) and Direct ML relative to Stefan recalibrated on the same labels. Error floor: covariate-conditional floor (Alaska 11.31, Lena Delta 14.83, Canada

17.90 cm) minus recalibrated Stefan RMSE; the Canadian line (-10.6 to -10.8 cm) lies below the axis. In Alaska with 1000 labels the residual model had 0.54 cm lower error; reductions with 500 and all labels (0.39 and 0.52 cm) did not hold under the common-resampling interval. No difference was established in the Lena Delta or Canada. (b) Alaska with all labels: error change by scoring block (25 splits summed) and the 343 label locations; blocks with under ten scored cells masked. (c) Error change split between and within ERA5-Land climate cells. In Alaska about 99% of the squared-error reduction came from the between-cell component (-1.03 cm; within-cell -0.01 cm); in the three-region mean, within-cell predictions increased error by 0.09 cm. (d) Recommended method per label interval: error change relative to the source-coefficient Stefan model, Lena Delta and Canada transfer pool (two-region mean of point estimates; 320 and 1000 labels not tested); placement and diagnosis steps below; selection from 40 labels rests on non-inferiority to a fixed recipe (Fig. 5c). (e) Workflow (d; placement reduced to random) in new random splits of the same regions versus random placement with the fixed recipe (-0.86 , -0.93 cm; 40, 160 labels) and versus recalibrated Stefan ($+0.08$, -1.35 , -1.51 cm; 10, 40, 160 labels); symbols, regions over 0.5 cm worse than recalibrated Stefan; designed after earlier results were viewed; re-test in reused regions, partly unblinded. Intervals: 95% block-bootstrap confidence intervals (10,000 resamples of 0.5° scoring blocks), weighted by cell and by block (key); ± 0.5 cm band, equivalence margin. Panels a to d were designed after the transfer results were viewed. Polar stereographic projection centred on Alaska, scale bar true at 70° N; coastlines, Natural Earth 50 m; Cartopy 0.23.0.

Figure 7. Spreading labels across blocks and covariate space lowered error versus random cells in Canada, not in the Lena Delta. (a to c) One draw of 100 labels in Canada (within-region test, split 1) by the three placement procedures named above the maps (Methods); small dots are candidate cells in the label blocks, and circle area is proportional to the labels selected per site (blocks closer than 40 km pooled). The maps illustrate the procedures and do not rank sites. (d) Error change relative to random cells for Anchor + residual ML (residual weight 0.25) against the number of labels within Canada, the Lena Delta and Alaska. (e) The same contrast in transfer with ten and forty labels. Intervals are 95% block-bootstrap confidence intervals (CIs; 10,000 resamples of 0.5° scoring blocks, conditional on the label draws), weighted by cell and by block as indicated in the key; the grey band marks ± 0.5 cm. In panel d, both spreading strategies lowered error in Canada by 2.5–3.2 cm at 50 and 100 labels and by 1.3–1.4 cm at 200 labels. In the Lena Delta, block stratification changed error by $+1.5$ to $+2.0$ cm at 20–500 labels, with higher error under both weightings at 100 labels ($+1.7$ cm; depends on the split-independence assumption). In Alaska the weightings disagreed at 50–200 labels (block-equal -1.1 to -1.7 cm, cell-weighted $+0.28$ to $+0.74$ cm). Variance-based active selection had 0.69 to 2.8 cm higher error at 50–200 labels in Alaska and the Lena Delta (one of six contrasts depends on the split-independence assumption) and smaller gains than spreading in Canada. In panel e, covariate spread lowered error in Canada at forty labels by 2.4 cm (95% CI 0.5 to 3.4 cm; depends on the split-independence assumption); at ten labels no region had higher error under both weightings, but the cell-weighted interval of block stratification in the Lena Delta lay above zero ($+1.9$ cm). The placement tests were

designed after earlier results had been viewed and registered before running. Polar stereographic projection centred on Canada, scale bar in panel a; coastlines, Natural Earth 50 m; Cartopy 0.23.0.

Table caption

Table 1. Target regions, labels and data sources. Labels is the number of label rows; 1 km locations groups the rows by a 0.009° cell index; 0.5° blocks are the units of the label and scoring split; Splits is the number of valid block splits in the transfer design; E_0 is the Stefan coefficient ($\text{cm } (^\circ\text{C d})^{-1/2}$) fitted on the source pool when the region is held out; Alaska share of source cells is the fraction of source cells located in Alaska. Main regions were used in earlier experiments and are retested here; Alaska is a reference target that is not averaged with them; new regions were added after the label tables had been fixed (licence-verified edition). Sub-regions, targets with point estimates only and the North Atlantic group are listed in Supplementary Table S11. Access conditions are given under Data availability.

Table 1 Target regions, labels and data sources.

Region	Label source	Labels	1 km locations	0.5° blocks	Splits	Source coefficient E_0 (cm per $\sqrt{(^{\circ}\text{C d})}$)	Alaska share of source cells (%)
<i>Main regions</i>							
Lena Delta	Points (ALLena)	3037	201	20	5	1.62	94
Canada	Points (ABoVE)	750	86	39	5	1.59	81
W Russia	CALM site means	31	29	21	5	1.59	78
E Russia	CALM site means	30	30	21	5	1.59	78
<i>Reference</i>							
Alaska	Points (ABoVE)	13,606	343	74	5	1.50	0
<i>New regions</i>							
Central Russia (expanded)	1 km cell means	57	57	13	5	1.59	78
Tibetan Plateau	1 km cell means	132	132	34	5	1.59	78

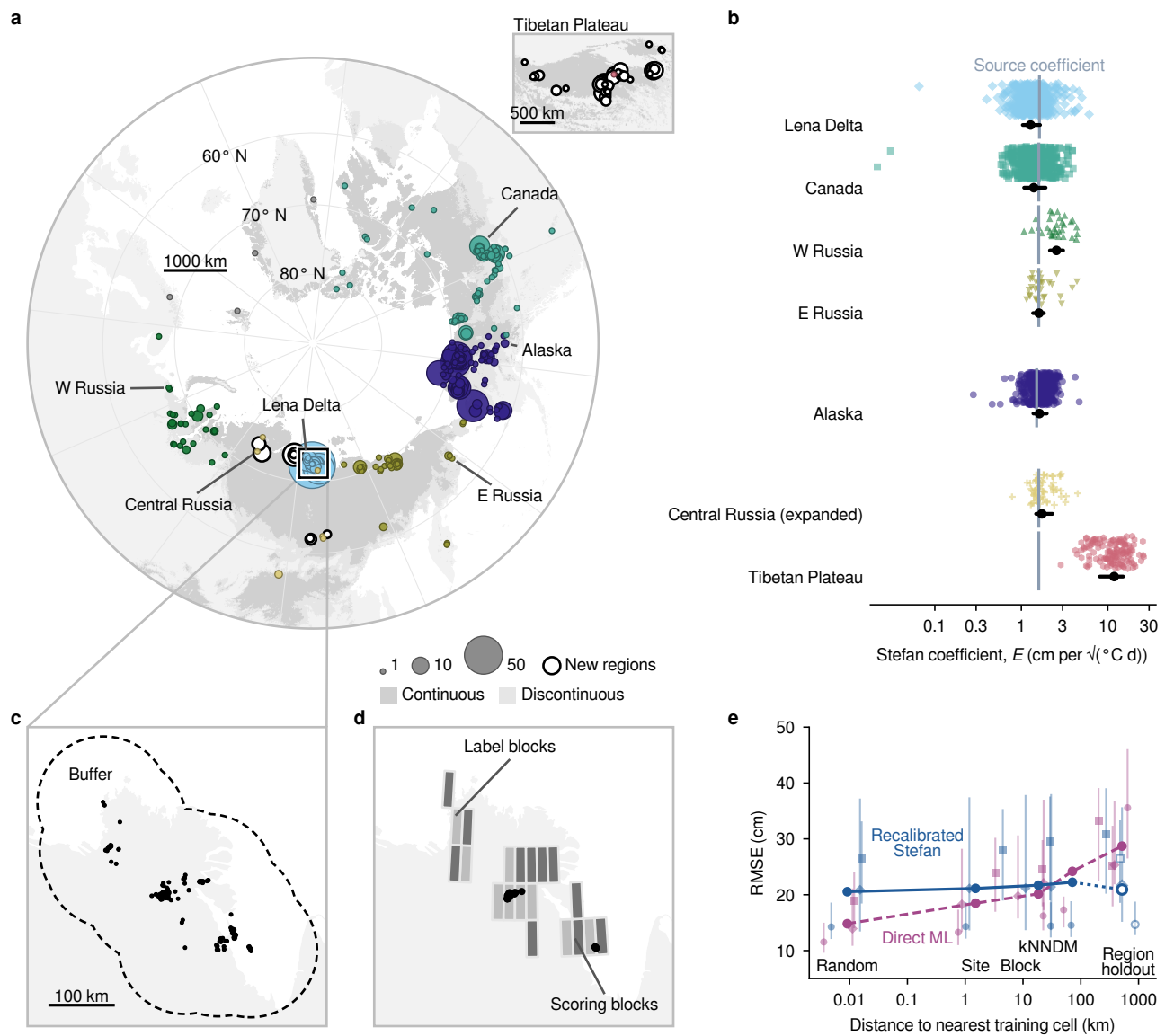


Figure 1

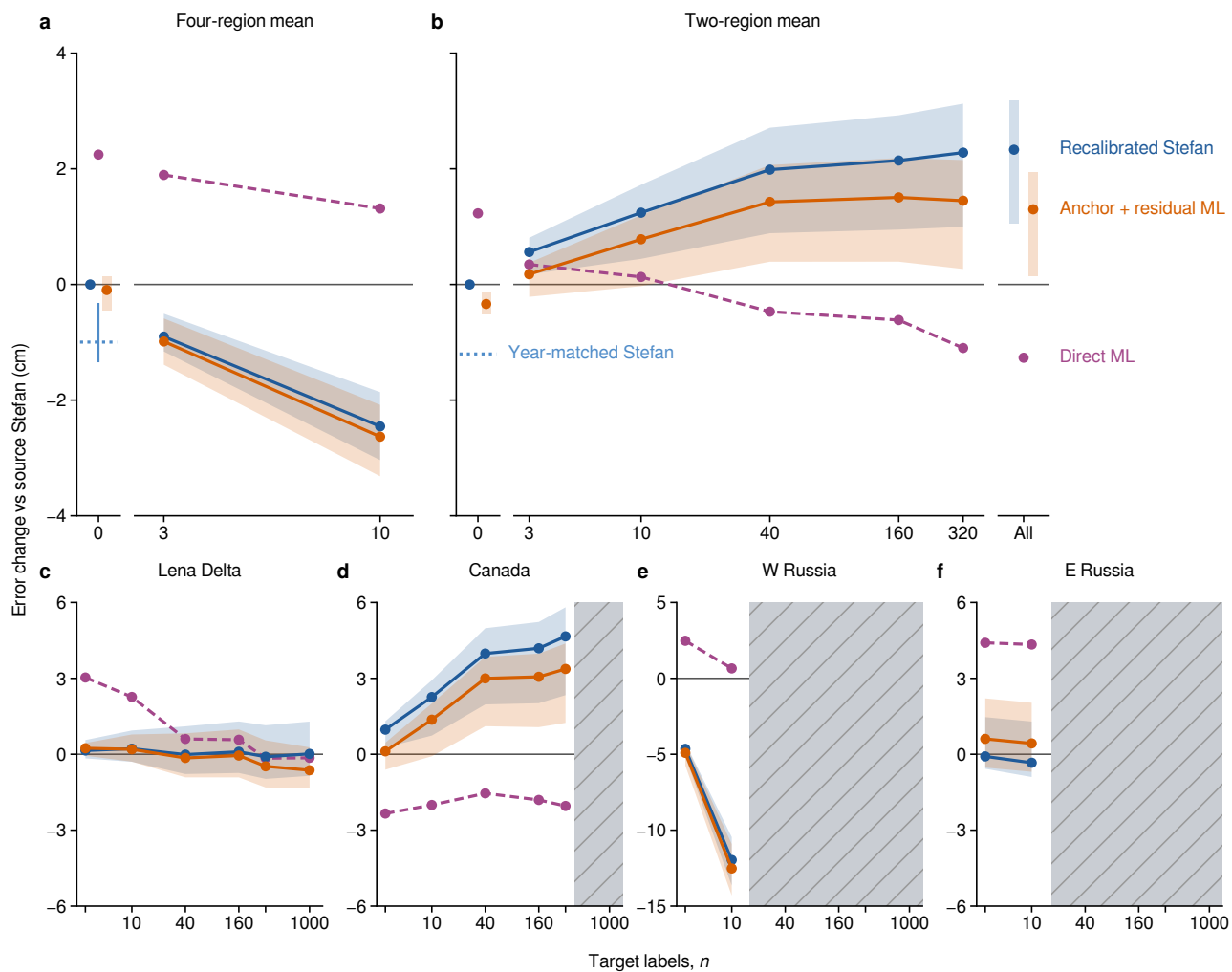


Figure 2

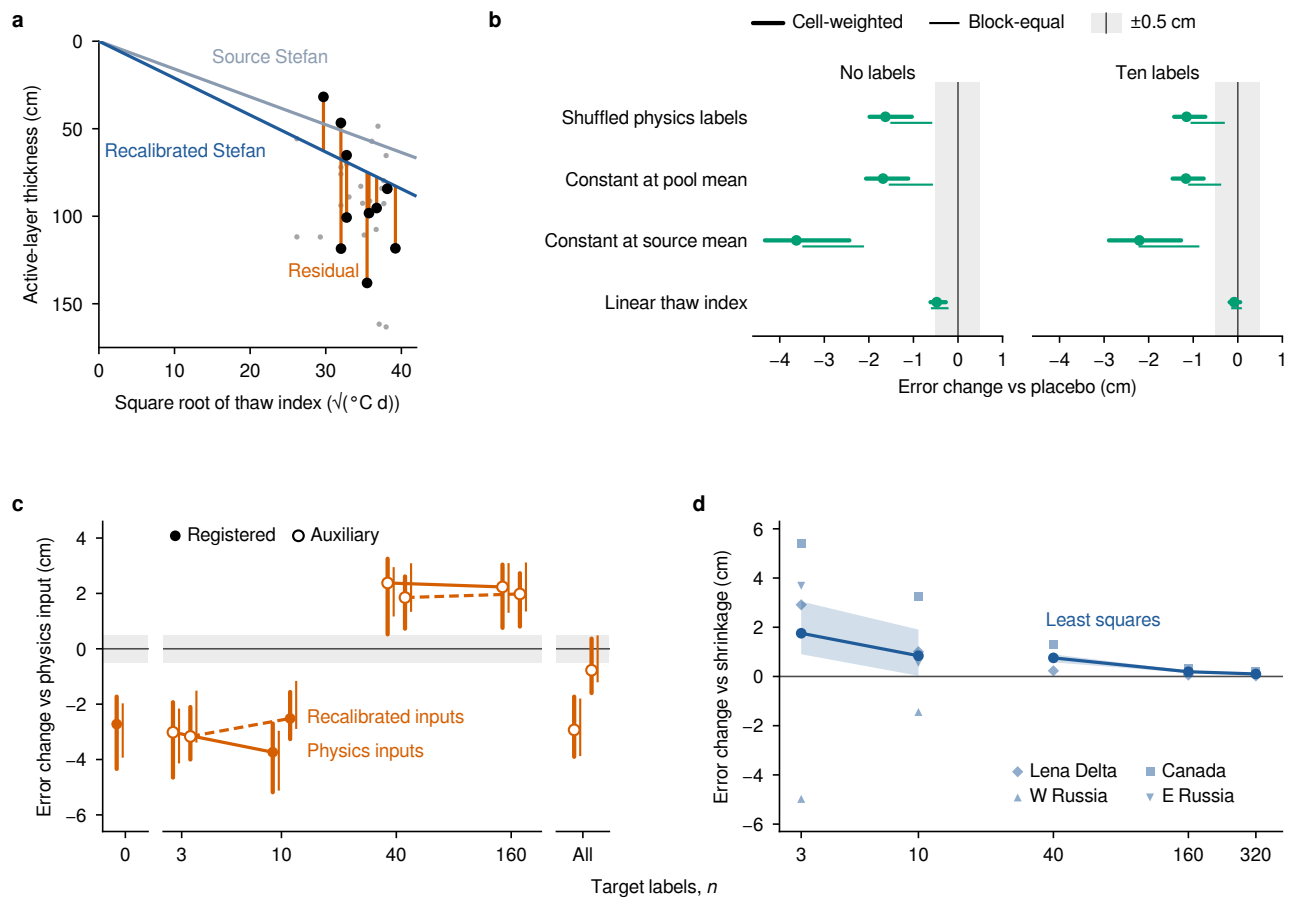


Figure 3

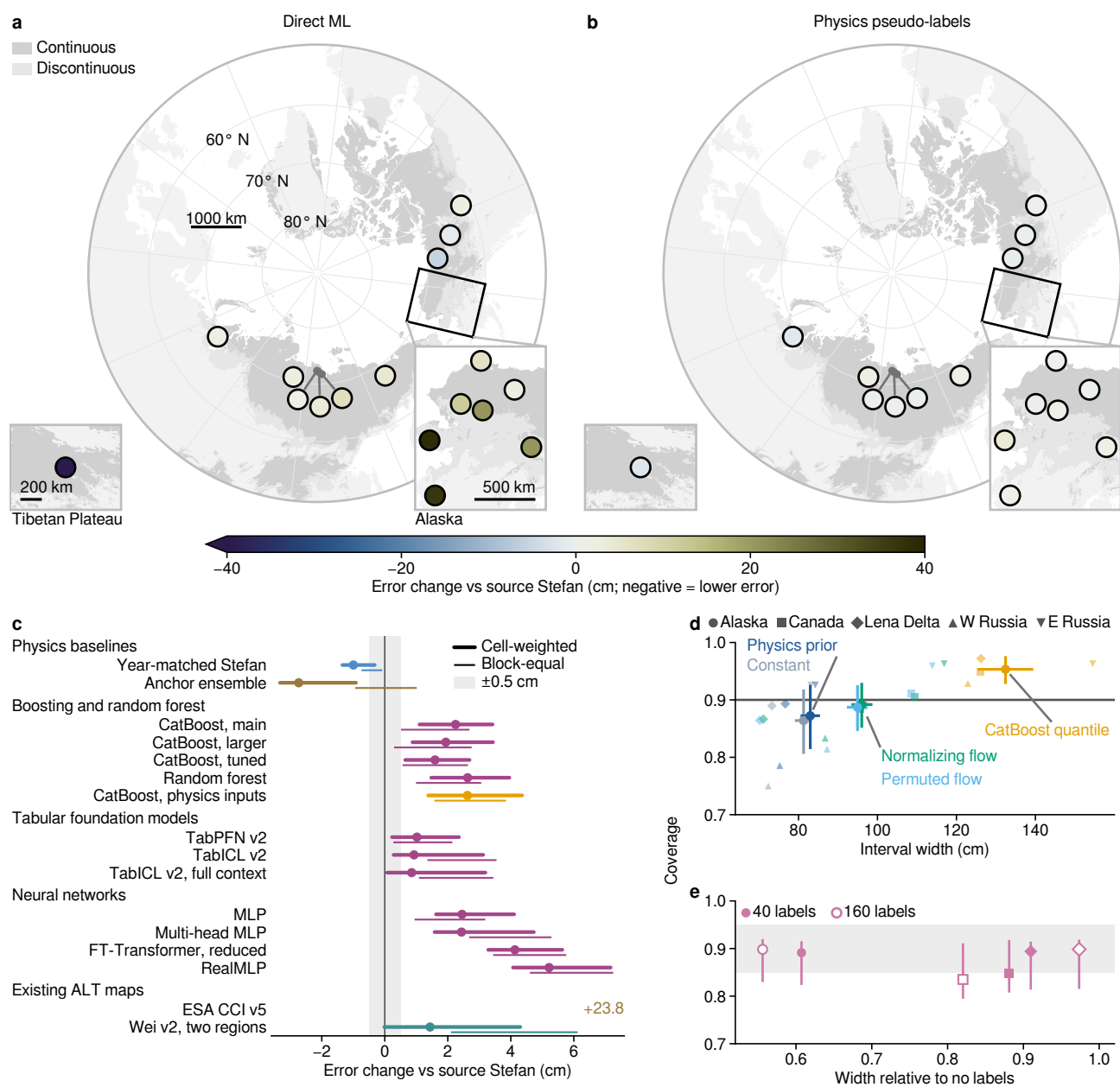


Figure 4

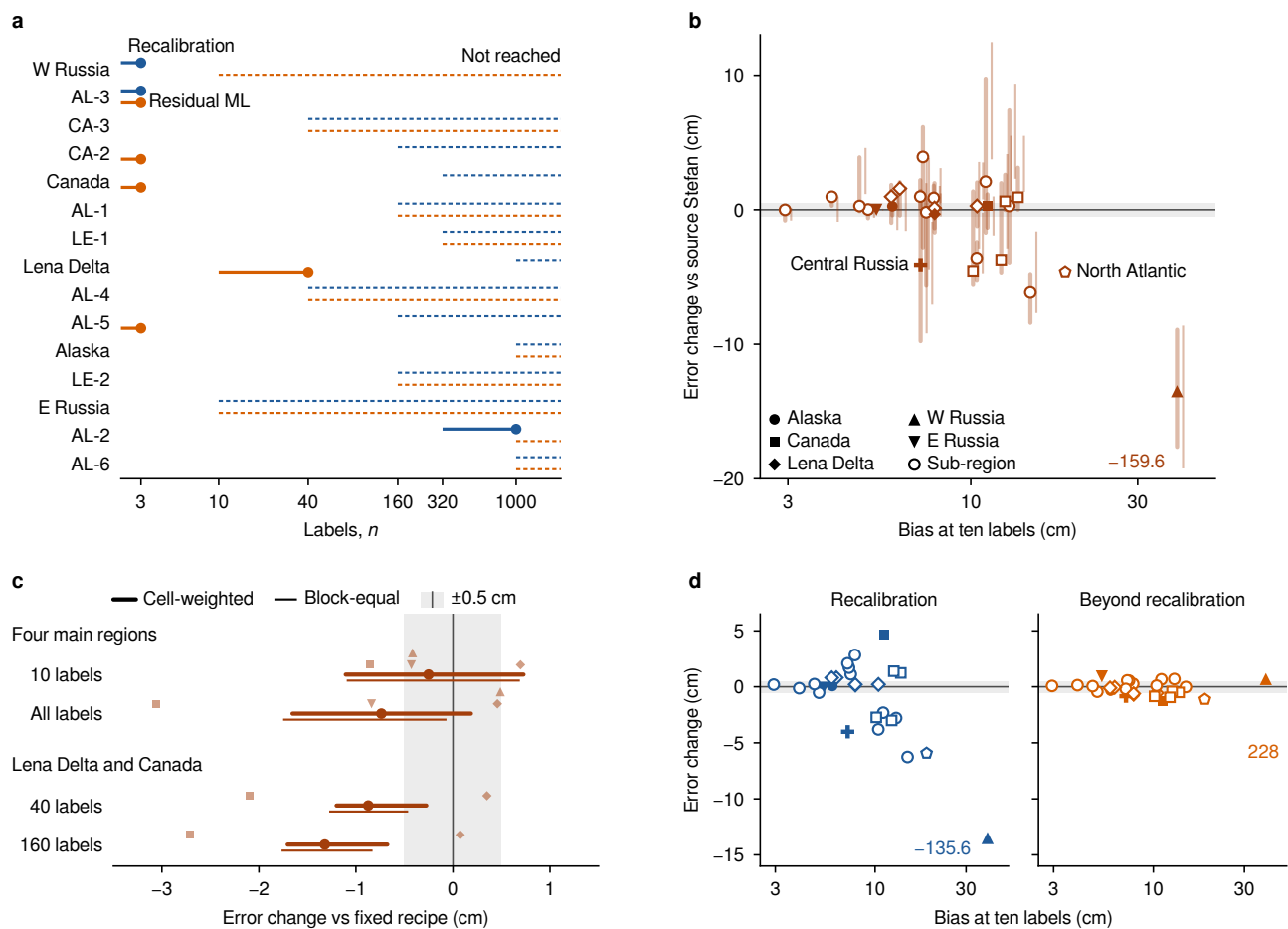


Figure 5

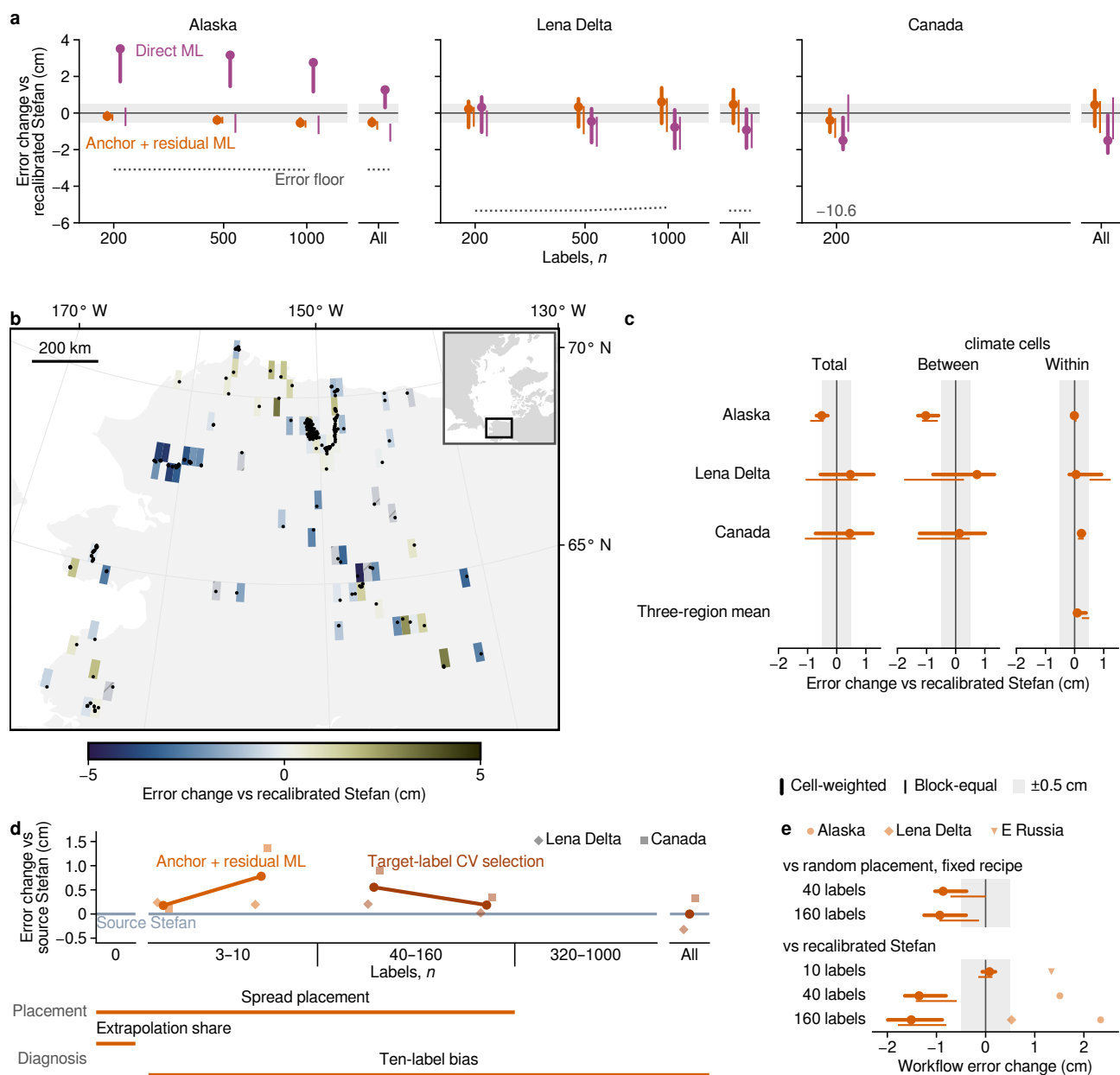


Figure 6

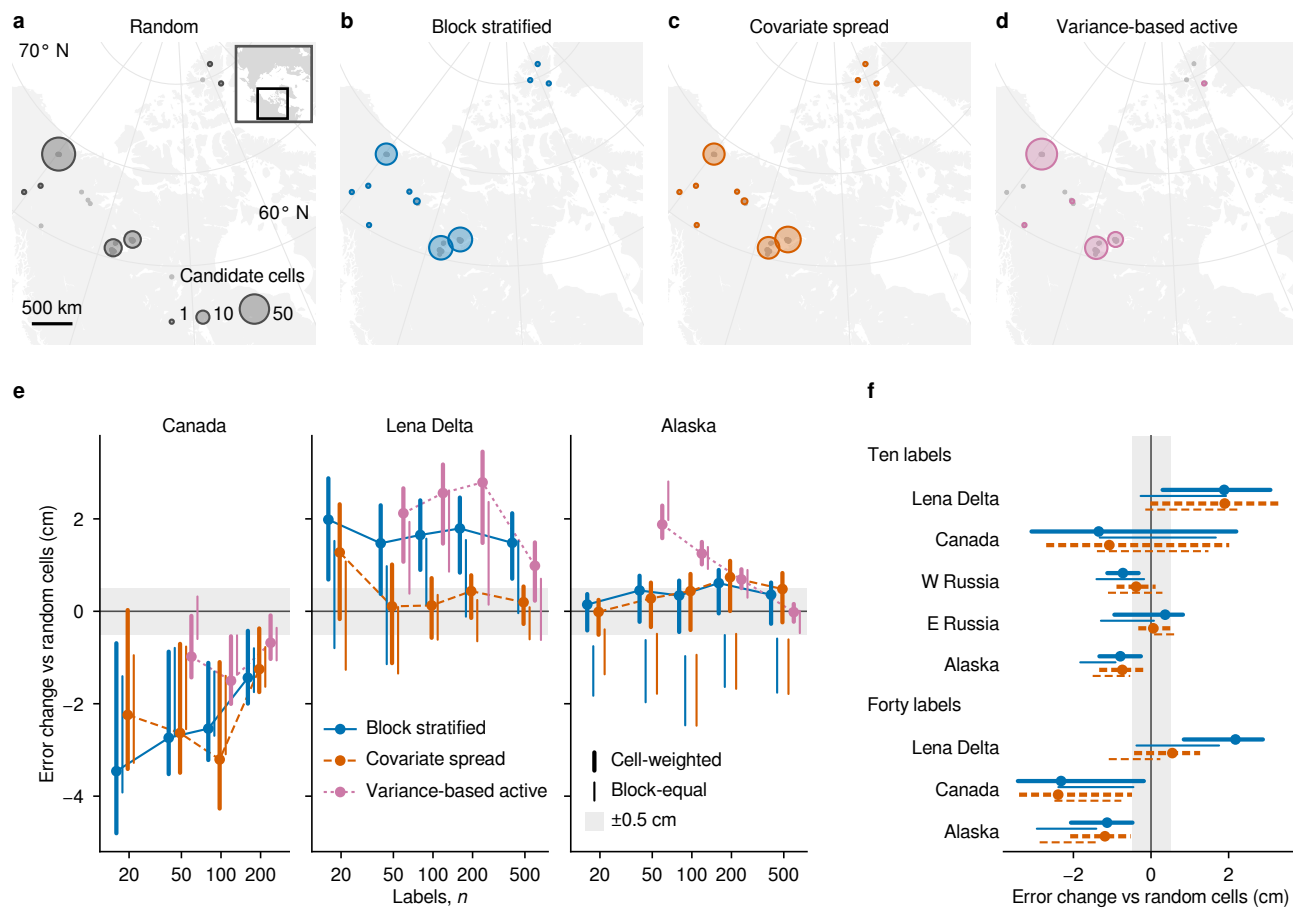


Figure 7